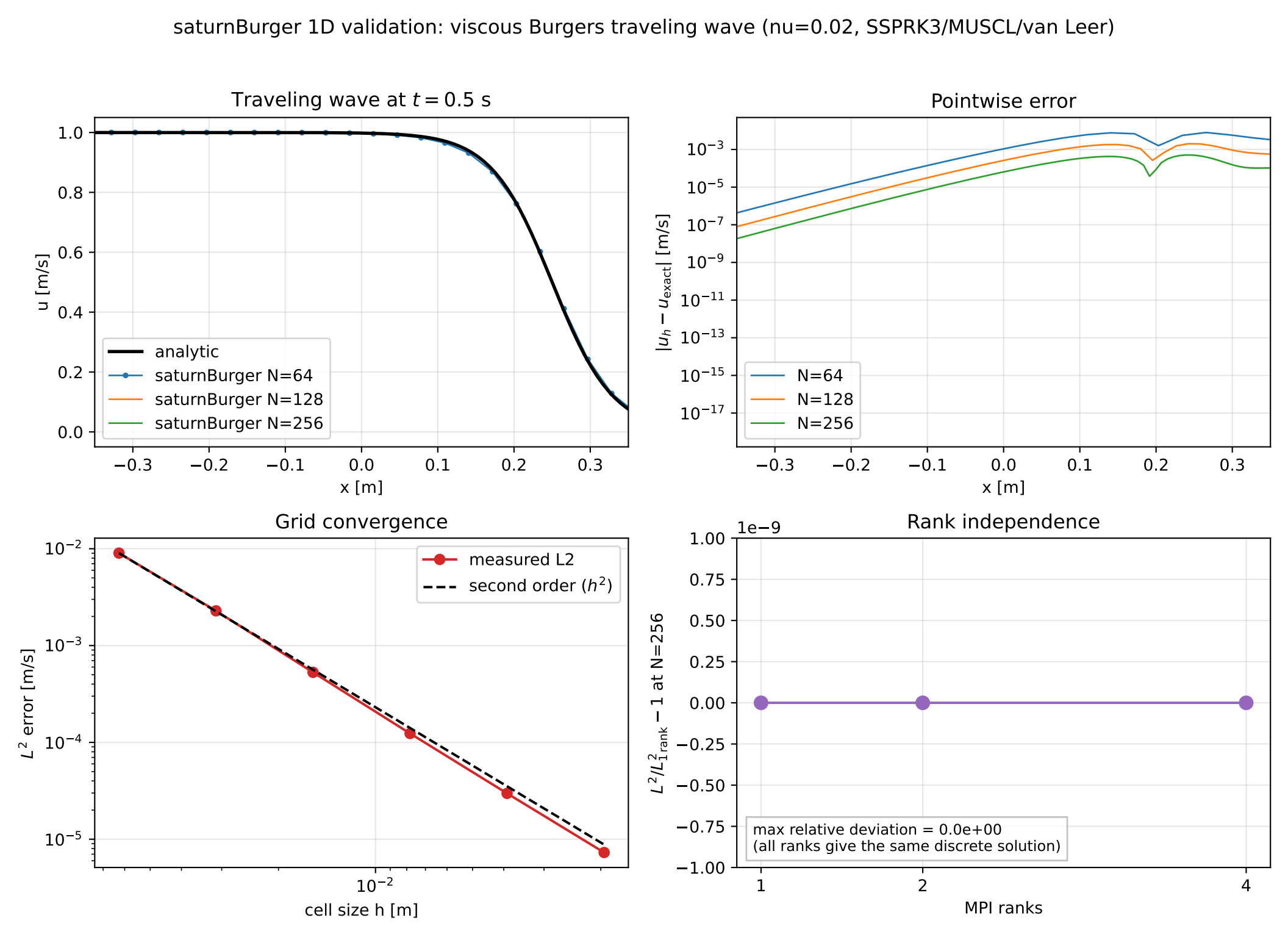
saturnBurger 1D validation: viscous Burgers traveling wave (nu=0.02, SSPRK3/MUSCL/van Leer)
Traveling wave at $t = 0.5$ s
u [m/s]
x [m]
analytic
saturnBurger N=64
saturnBurger N=128
saturnBurger N=256
Pointwise error
$|u_h - u_{exact}|$ [m/s]
x [m]
N=64
N=128
N=256
Grid convergence
L^2 error [m/s]
cell size h [m]
measured L2
second order (h^2)
Rank independence
$L^2/L^2_{1\,rank} - 1$ at N=256
1e−9
MPI ranks
max relative deviation = 0.0e+00
(all ranks give the same discrete solution)